



Top 30 SOC Analyst (L1) Interview Questions & Answers

1. What is a SOC?

A **Security Operations Center (SOC)** is a centralized unit that monitors, detects, analyzes, and responds to cybersecurity threats in real-time.

2. What are the roles and responsibilities of a SOC Analyst?

A SOC Analyst is responsible for monitoring security alerts, investigating incidents, analyzing logs, reporting threats, and responding to cyberattacks.

3. What is SIEM?

Security Information and Event Management (SIEM) is a tool that collects, analyzes, and correlates security logs from different sources to detect threats.

4. Name some popular SIEM tools.

Some popular SIEM tools are **Splunk, IBM QRadar, ArcSight, Microsoft Sentinel, and LogRhythm.**

5. What is the difference between IDS and IPS?

- **IDS (Intrusion Detection System)** only detects malicious activity.
- **IPS (Intrusion Prevention System)** detects and blocks threats.

6. What is a Firewall?

A firewall is a security device that monitors and controls incoming and outgoing network traffic based on security rules.

7. What is a false positive and false negative in cybersecurity?

- **False Positive:** A legitimate activity is mistakenly flagged as malicious.
- **False Negative:** A real attack goes undetected.

8. What is Phishing?

Phishing is a cyber attack where attackers trick users into revealing sensitive information (like passwords) using fake emails or websites.

9. What is Malware?

Malware (malicious software) includes viruses, worms, Trojans, ransomware, and spyware that harm systems or steal data.

10. What is a DDoS attack?

A **Distributed Denial of Service (DDoS) attack** floods a server or network with massive traffic to make it unavailable.

11. What are the different types of security incidents?

A. Security incidents include malware infections, phishing, data breaches, insider threats, and unauthorized access.

12. How do you analyze security logs?

By using **SIEM tools**, checking for anomalies, reviewing IP addresses, timestamps, and identifying unusual activities.

13. What is the MITRE ATT&CK framework?

A knowledge base of adversary tactics, techniques, and procedures (TTPs) used to understand attack patterns.

14. What is Threat Intelligence?

Threat Intelligence provides information about cyber threats, helping analysts detect and prevent attacks proactively.

15. What are Indicators of Compromise (IoCs)?

IoCs are signs of a cyber attack, such as **unusual IP addresses, file hash changes, strange login attempts, or suspicious traffic**.

16. What is an Endpoint Detection and Response (EDR) tool?

EDR tools **monitor, detect, and respond** to security threats at endpoints (laptops, servers, etc.). Examples: **CrowdStrike, Microsoft Defender, SentinelOne**.

17. What is Privilege Escalation?

It's an attack where a hacker gains **higher-level access (admin/root)** to a system by exploiting vulnerabilities.

18. What is Ransomware?

Ransomware is a malware that **encrypts files and demands a ransom** to restore access.

19. What is the difference between a virus and a worm?

- **Virus:** Needs a host file to spread.
- **Worm:** Spreads itself across networks without a host file.

20. What are the different types of cyber attacks?

- Phishing
- Malware
- Ransomware
- Man-in-the-Middle (MitM)
- SQL Injection
- DDoS Attack

21. What is DNS Spoofing?

DNS Spoofing tricks users into visiting fake websites by altering the domain name system (DNS) records.

22. What is Social Engineering?

A technique where attackers manipulate people into revealing confidential information (e.g., phishing, pretexting, baiting).

23. How do you investigate a security incident?

- Identify the alert
- Analyze logs
- Check affected systems
- Verify IoCs
- Take action (block IP, isolate system, report)

24. What is the CIA Triad?

The three pillars of cybersecurity:

- **Confidentiality** (protecting data)
- **Integrity** (ensuring data is not altered)
- **Availability** (making data accessible)

25. What is an Insider Threat?

A security risk caused by employees, contractors, or partners who misuse their access to harm an organization.

26. What is Penetration Testing?

A simulated cyberattack conducted by ethical hackers to find vulnerabilities in a system before real attackers do.

27. What is Honeypot?

A **decoy system** designed to attract hackers and analyze their attack methods.

28. What is a Zero-Day Attack?

An attack that exploits a vulnerability before a patch is available.

29. How do you stay updated with cybersecurity trends?

Follow cybersecurity blogs, attend webinars, take online courses, and use platforms like **KrebsOnSecurity**, **Dark Reading**, and **Cybrary**.

30. What steps should be taken after detecting a security breach?

- Isolate affected systems
- Analyze the attack
- Identify the root cause
- Apply patches/fixes
- Report the incident
- Strengthen security



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